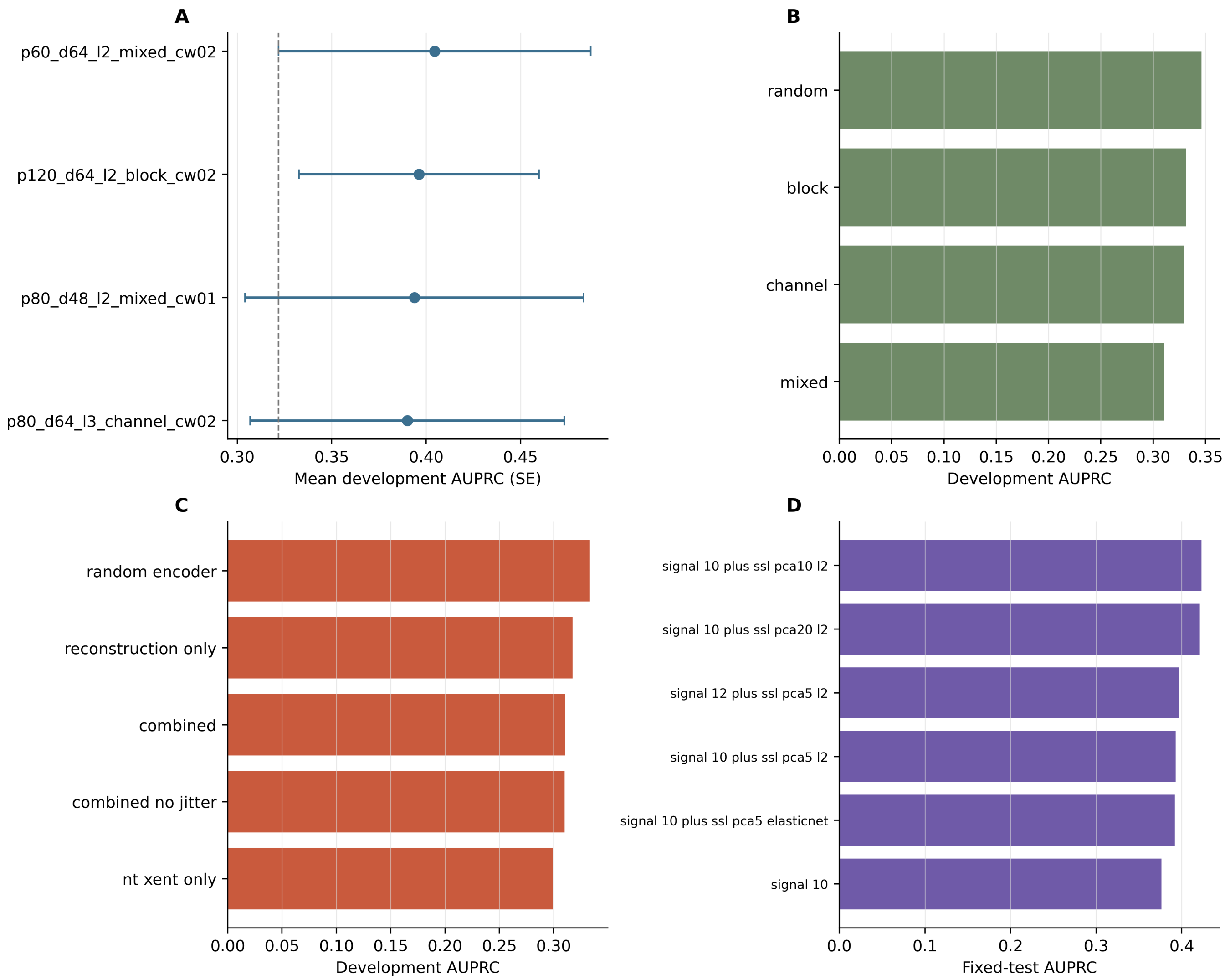
Supplementary material

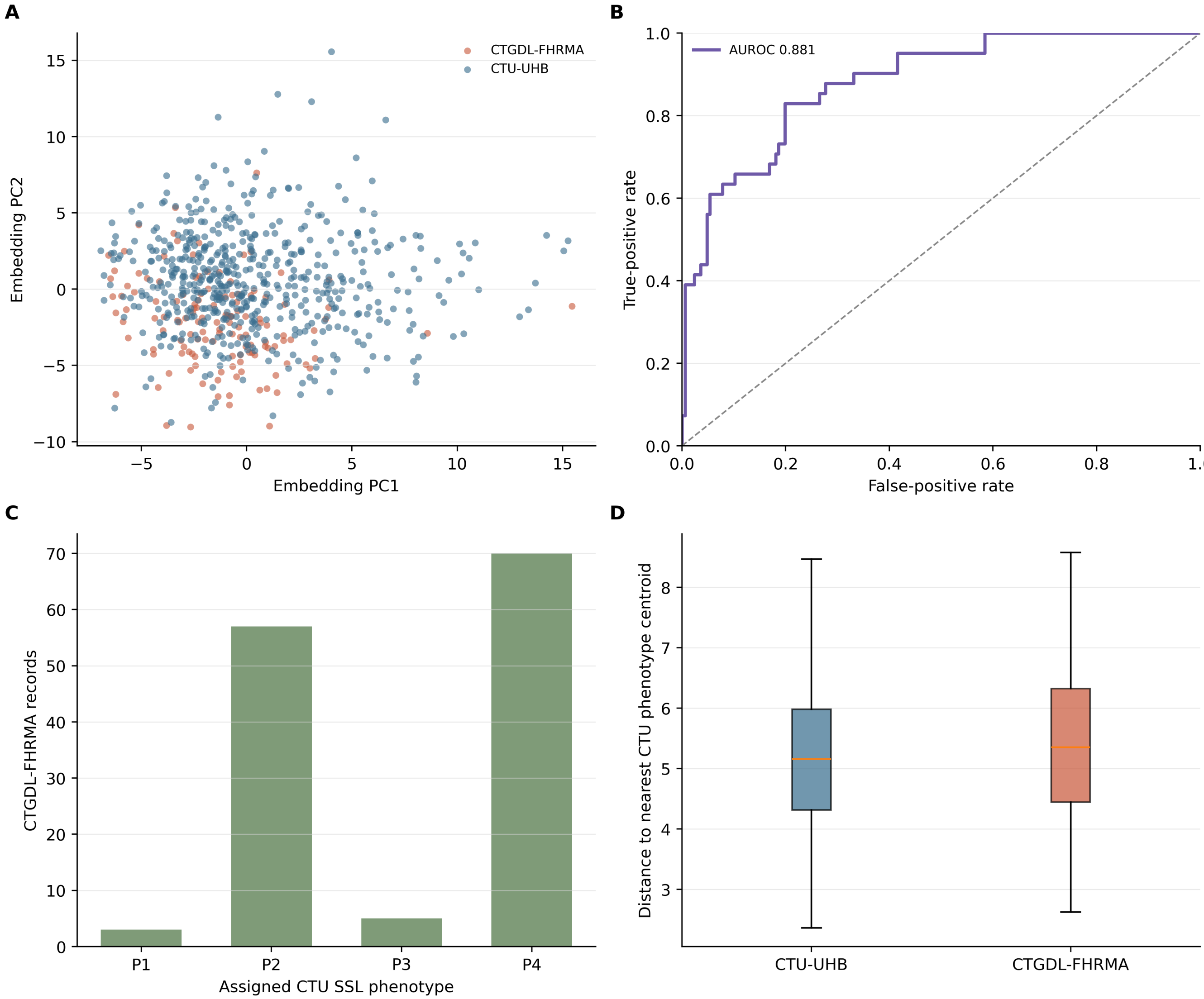
Self-supervised biomedical signal representation learning from intrapartum cardiotocography: stability, ablation, and exploratory neonatal risk enrichment

Supplementary Figure S1



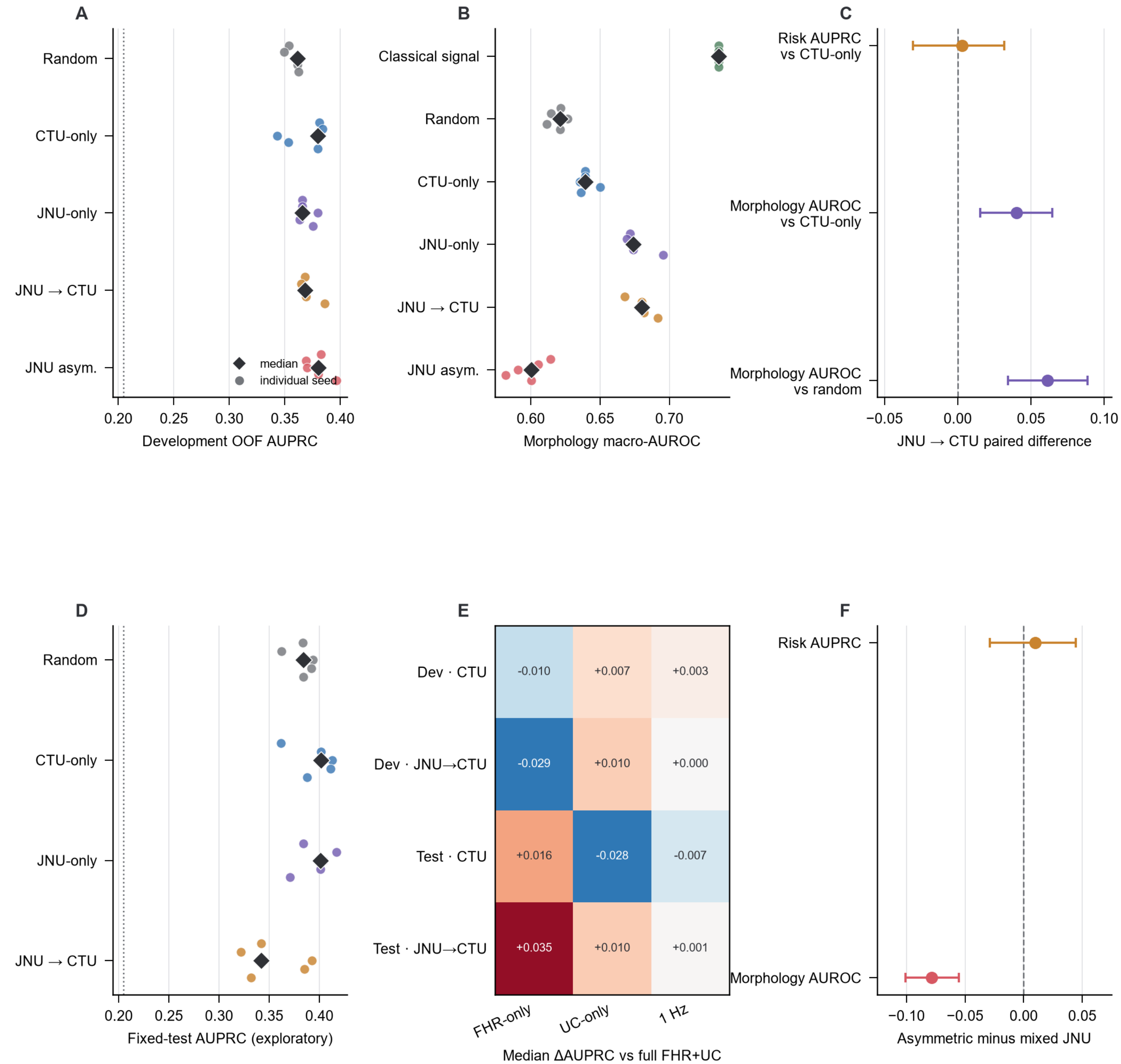
AI model selection and sensitivity. (A) Mean development AUPRC and standard error for four pre-specified architectures. The vertical dashed line is the one-standard-error eligibility threshold; p80-d48-l2 mixed was the smallest eligible model. (B) Mask-strategy sensitivity on a fixed development split. (C) Objective ablation including the random-encoder negative control. (D) Fixed-test dimensionality, regularization and missingness sensitivities. Panels B and C were not used to reselect the final architecture; Panel D is exploratory because the fixed test set had prior exposure.

Supplementary Figure S2



External representation transport and domain shift. (A) Principal-component projection of selected-encoder embeddings from CTU-UHB and CTGDL-FHRMA. (B) Domain-classifier ROC curve (AUROC 0.8811). (C) CTGDL-FHRMA assignments to CTU-fitted SSL phenotype centroids. (D) Distance to the nearest CTU phenotype centroid. This is representation/domain-shift validation, not external neonatal-outcome validation.

Supplementary Figure S3



Public JNU-CTG cross-domain pretraining sensitivity. (A) Five-seed development out-of-fold AUPRC by encoder origin. The dotted line denotes development outcome prevalence. (B) Macro-AUROC across seven CTU expert-morphology probes; classical signal features are a supervised conventional-feature comparator. (C) Paired hierarchical bootstrap differences for JNU pretraining followed by CTU adaptation. JNU transfer improved morphology probes relative to CTU-only and random embeddings, whereas its risk-enrichment interval crossed zero. (D) Exploratory fixed-test AUPRC after the development promotion decision; the dotted line denotes test outcome prevalence. (E) Median AUPRC change for FHR-only, UC-only and 1-Hz inputs relative to full 4-Hz FHR+UC. Directions differed between development and fixed-test analyses. (F) Channel-asymmetric minus mixed JNU masking. The morphology interval was negative and the risk interval crossed zero. Fixed-test panels retain potential selection-induced optimism; no JNU result represents external outcome validation.

Supplementary Table S1. Fixed cohort split

Split	Records	Events	Prevalence
Development	386	79	0.2047
Fixed test	166	34	0.2048

Supplementary Table S2. Development-only architecture selection

Architecture	Mean AUPRC	SE	Parameters	Selected
p60-d64-l2 mixed cw0.2	0.4045	0.0826	121,532	No
p120-d64-l2 block cw0.2	0.3962	0.0636	126,712	No
p80-d48-l2 mixed cw0.1	0.3938	0.0897	74,768	Yes
p80-d64-l3 channel cw0.2	0.3900	0.0832	172,816	No

The highest mean AUPRC was 0.4045; its one-standard-error threshold was 0.3218. The smallest eligible architecture was selected. Fold-level results and split assignments are provided as CSV files.

Supplementary Table S3. Development-only objective and mask sensitivities

Objective	AUPRC	AUROC	Brier
Random encoder	0.3333	0.6883	0.2385
Reconstruction only	0.3174	0.6656	0.2408
NT-Xent only	0.2991	0.6695	0.2433
Reconstruction + NT-Xent	0.3106	0.6760	0.2448
Combined, no jitter	0.3100	0.6786	0.2453

Mask strategy	AUPRC	AUROC	Brier
Random	0.3461	0.7104	0.2300
Block	0.3312	0.7039	0.2372
Channel	0.3296	0.6818	0.2352
Mixed	0.3106	0.6760	0.2448

All results used the same fixed 289/97 development split. They are component sensitivities, not test-set selection results.

Supplementary Table S4. Five-seed stability and paired AUPRC differences

Seed	AUPRC	AUROC	Brier	ΔAUPRC	95% CI
20260521	0.3929	0.5996	0.2636	0.0164	−0.0596, 0.0763
20260522	0.3935	0.5867	0.2628	0.0170	−0.0623, 0.0864
20260523	0.4042	0.6058	0.2628	0.0277	−0.0597, 0.1013
20260524	0.4021	0.5934	0.2623	0.0257	−0.0551, 0.0886
20260525	0.3916	0.5954	0.2643	0.0151	−0.0665, 0.0880

The signal baseline was fixed across seeds (AUPRC 0.3765, AUROC 0.6428, Brier 0.2482). Differences are reduced SSL minus signal baseline using 2,000 paired bootstrap resamples.

Supplementary Table S5. Dimensionality, regularization and missingness sensitivities

Model	Predictors	EPV	AUPRC	AUROC
Signal 10	10	7.90	0.3765	0.6428
Signal 10 + SSL PCA5, L2	15	5.27	0.3929	0.5996
Signal 10 + SSL PCA10, L2	20	3.95	0.4229	0.6085
Signal 10 + SSL PCA20, L2	30	2.63	0.4210	0.6179
Signal 10 + SSL PCA5, elastic net	15	5.27	0.3918	0.6005
Signal 12 + SSL PCA5, L2	17	4.65	0.3967	0.6217

EPV, events per predictor, uses 79 development events. PCA10/20 and missingness results were not used to replace the prespecified PCA5 primary model.

Supplementary Table S6. Phenotype incremental sensitivity

k	Model	AUPRC	AUROC	Predictors
3	Reduced SSL	0.3929	0.5996	15
3	+ signal phenotype	0.3796	0.5762	17
3	+ SSL phenotype	0.3804	0.6005	17
4	Reduced SSL	0.3929	0.5996	15
4	+ signal phenotype	0.3978	0.6047	18
4	+ SSL phenotype	0.3511	0.5858	18
4	+ both phenotypes	0.3457	0.5831	21
5	Reduced SSL	0.3929	0.5996	15
5	+ signal phenotype	0.3944	0.5960	19
5	+ SSL phenotype	0.3736	0.5882	19

Complete k = 3, 4 and 5 rows, including both-phenotype models, are supplied as CSV. Phenotypes are secondary interpretation variables.

Supplementary Table S7. Endpoint sensitivity

Outcome	Model	Dev events	Test events	AUPRC	AUROC
Composite	Signal	79	34	0.3765	0.6428
Composite	Reduced SSL	79	34	0.3929	0.5996
pH <7.15	Signal	72	33	0.4161	0.6437
pH <7.15	Reduced SSL	72	33	0.4044	0.6138
Apgar5 <7	Signal	15	4	0.0593	0.4599
Apgar5 <7	Reduced SSL	15	4	0.0591	0.6173
pH <7.10	Signal	37	19	0.4507	0.7479
pH <7.10	Reduced SSL	37	19	0.4588	0.7429
pH <7.05	Signal	25	15	0.5721	0.8283
pH <7.05	Reduced SSL	25	15	0.5382	0.7991

The Apgar-only analysis has fewer than 10 test events and is reported only as an unstable exploratory estimate.

Supplementary Table S8. Calibration metrics

Model	Calibration	Brier	ECE	Intercept	Slope
Signal	Raw	0.2482	0.2874	−1.4267	0.6518
Signal	Platt	0.1543	0.0386	−0.2404	0.9006
Signal	Isotonic	0.1581	0.0588	−0.5425	0.6506
Reduced SSL	Raw	0.2636	0.2747	−1.3920	0.3860
Reduced SSL	Platt	0.1579	0.0915	−0.5638	0.6440
Reduced SSL	Isotonic	0.1694	0.0878	−0.8558	0.4171

Supplementary Table S9. Historical exploratory result

Historical model	AUPRC	AUROC	Status
Signal + 64 SSL dimensions + signal phenotype	0.4016	0.6168	Exploratory only. The fixed test set contributed to configuration ranking, so this estimate is not independent validation and is excluded from revised model selection and primary reporting.

Supplementary Table S10. Selected encoder and computing environment

Parameter	Value	Parameter	Value
Patch length	80	Model dimension	48
Layers	2	Attention heads	4
Mask strategy	Mixed	Mask fraction	0.25
Reconstruction weight	1.0	Contrastive weight	0.1
Temperature	0.2	Jitter SD	0.02
Epochs	4	Batch size	64
Learning rate	0.001	Weight decay	0.0001
Parameters	74,768	Training windows	5,301
Framework	PyTorch 2.12.0+cu130	Device	RTX 5080 Laptop GPU

Supplementary Table S11. External representation analysis

Metric	Value
CTU-UHB records	552
CTGDL-FHRMA records	135
Embedding dimension	48
Domain-classifier AUROC	0.8811
External outcome validation	No
Representation/domain-shift analysis	Yes

Supplementary Tables S12–S13. Reproducibility audit

Supplementary Table S12 provides SHA256 hashes for locked inputs. Supplementary Table S13 provides the 23-item output cross-audit, including fold-membership checks, recomputation of AUPRC, AUROC and Brier scores from record-level probability tables, seed completeness, endpoint event counts and figure-file checks. Both are supplied as machine-readable CSV files. The audit status was PASS.

Supplementary Table S14. Encoder-origin risk stability

Encoder origin	Dev AUPRC	Dev AUROC	Fixed-test AUPRC	Fixed-test AUROC
Random initialization	0.3611	0.6584	0.3840	0.5978
CTU-development-only SSL	0.3797	0.6691	0.4017	0.5945
JNU-only SSL	0.3658	0.6702	0.4012	0.6092
JNU pretraining $\rightarrow$ CTU adaptation	0.3682	0.6720	0.3420	0.5951
JNU channel-asymmetric sensitivity	0.3802	0.6475	–	–

Values are medians across seeds 20260521–20260525. Development results are three-fold out-of-fold estimates. Fixed-test results are exploratory because the test set had historical exposure. The channel-asymmetric route was development-only.

Supplementary Table S15. Frozen-encoder morphology probes

Representation	Macro-AUPRC	Macro-AUROC
Classical signal features	0.5666	0.7357
Random encoder	0.4543	0.6211
CTU-development-only SSL	0.4789	0.6392
JNU-only SSL	0.5155	0.6738
JNU pretraining → CTU adaptation	0.5076	0.6802
JNU channel-asymmetric sensitivity	0.4460	0.6005

Values are medians across five seeds and seven expert-event tasks. Probes were fitted and evaluated only within record-grouped CTU development folds.

Supplementary Table S16. JNU paired hierarchical bootstrap comparisons

Data role	Metric	Contrast	Difference	95% CI	Positive seeds
Development	Risk AUPRC	JNU→CTU vs CTU-only	0.0030	−0.0304, 0.0316	3/5
Development	Morphology AUROC	JNU→CTU vs CTU-only	0.0401	0.0155, 0.0647	5/5
Development	Morphology AUROC	JNU→CTU vs random	0.0611	0.0344, 0.0884	5/5
Development	Morphology AUROC	JNU→CTU vs signal	−0.0555	−0.0924, −0.0200	0/5
Development	Risk AUPRC	Asymmetric vs mixed JNU	0.0098	−0.0290, 0.0444	4/5
Development	Morphology AUROC	Asymmetric vs mixed JNU	−0.0787	−0.1006, −0.0552	0/5
Fixed test	Risk AUPRC	JNU→CTU vs CTU-only	−0.0403	−0.0766, 0.0016	0/5
Fixed test	Risk AUPRC	JNU-only vs CTU-only	0.0005	−0.0204, 0.0240	2/5

Each interval used 2,000 paired hierarchical bootstrap resamples. Fixed-test rows are exploratory and were not used for promotion.

Supplementary Table S17. Channel and sampling-rate sensitivity

Data role	Encoder	Input	Median AUPRC	Δ vs full
Development	CTU-only	FHR-only	0.3645	−0.0153
Development	CTU-only	UC-only	0.3600	−0.0197
Development	CTU-only	1 Hz	0.3676	−0.0122
Development	JNU→CTU	FHR-only	0.3405	−0.0276
Development	JNU→CTU	UC-only	0.3789	0.0107
Development	JNU→CTU	1 Hz	0.3660	−0.0022
Fixed test	CTU-only	FHR-only	0.4113	0.0096
Fixed test	CTU-only	UC-only	0.3601	−0.0417
Fixed test	CTU-only	1 Hz	0.3910	−0.0107
Fixed test	JNU→CTU	FHR-only	0.3961	0.0541
Fixed test	JNU→CTU	UC-only	0.3521	0.0101
Fixed test	JNU→CTU	1 Hz	0.3323	−0.0097

Values are five-seed medians. The direction reversal across data roles precludes selecting an input variant from these sensitivities.

Supplementary Table S18. JNU-CTG data and leakage audit

Audit item	Result
Archive identifier	Zenodo 21800730; CC BY 4.0
Archive MD5	`ac1cfcba2f1b3336596544211d776771`; matched
Waveform readability	20,769/20,769 records readable (100%)
Patient grouping	12,606 unique patient groups retained
SSL windows	62,307 non-overlapping 10-minute windows; FHR+UC; 4 Hz
JNU labels used in SSL	None; Apgar, asphyxia, FIGO, demographics and other clinical labels withheld
CTU fold isolation	Fold-specific adaptation used development-training records only
Fixed-test access	Opened only after the development promotion decision; historical exposure disclosed
Promotion decision	Supplement-only because the risk-enrichment gate failed
External outcome validation	Not performed